

Olympiad Test Paper — Mathematics (Class 7)

Chapter 15: Finding the Unknown

Subject	Mathematics (NCERT Class 7)
Chapter	15 — Finding the Unknown
Total Questions	60 (Multiple Choice, single correct)
Maximum Marks	240
Suggested Time	90 minutes

Marking scheme: +4 for each correct answer · -1 for each wrong answer · 0 if unattempted.

Instructions: Each question has four options (A), (B), (C), (D); exactly one is correct. Every item reduces to ONE linear equation in one unknown, but the modelling is layered: deeply nested brackets, fractions-of-remainders, multi-condition age/digit/mixture set-ups, and solve-then-evaluate questions where the asked quantity is NOT x itself — watch sign flips inside nested braces, never average two rates, and re-read what is actually asked. Do not write on this paper — mark answers on your answer sheet.

1. Solve: $5[2x - \{3 - (x - 4)\}] - 7 = 8x + 14$.

- (A) 6 (B) 8
(C) -8 (D) 4

2. Solve: $9x - 4[2x - \{5 - (x - 2)\}] = 3x - 8$.

- (A) 6 (B) 12
(C) -6 (D) 4

3. Solve: $3(2x - 1) - 2[x - (3 - 2x)] = x + 5$.

- (A) 2 (B) 1
(C) 8 (D) -2

4. A number is multiplied by 4 and 9 is subtracted; the result equals the same number doubled and increased by 7. The number is:

- (A) -8 (B) 1
(C) 16 (D) 8

5. Solve: $(3x - 1)/4 - (2x + 3)/6 = (x + 11)/12$.

- (A) -5 (B) 3
(C) 7 (D) 5

6. A man spends $1/4$ of his money, then $1/3$ of what remains, and is left with ₹360. His original amount (in ₹) was:

- (A) 540 (B) 720
(C) 480 (D) 900

7. Three consecutive integers are such that the sum of the first and the third is 4 more than the second. The middle integer is:

- (A) 3 (B) 5
(C) 2 (D) 4

8. The present ages of a father and son add up to 60 years. In 12 years the father will be twice as old as the son will be then. The son's present age (in years) is:

- (A) 44 (B) 16
(C) 20 (D) 12

9. Solve: $(3x - 1)/3 - (x + 2)/6 = (6x - 10)/6$.

- (A) -6 (B) 3
(C) 2 (D) 6

10. A cyclist covers a distance at 12 km/h and returns over the same road at 18 km/h. The whole trip takes 5 hours. The one-way distance (in km) is:

- (A) 75 (B) 30
(C) 36 (D) 45

11. A mother is now 4 times as old as her daughter. Eight years ago she was 8 times as old. The mother's present age (in years) is:

- (A) 14 (B) 48
(C) 56 (D) 64

12. Solve: $(4x - 1)/3 - (5x - 1)/6 = (4x - 9)/6$.

- (A) -8 (B) 8
(C) 4 (D) 2

13. Two-thirds of a number is 4 more than half of the number. The number is:

- (A) 24 (B) 48
(C) 12 (D) 8

14. Solve: $(4x + 1)/4 - (3x - 3)/8 = (6x + 7)/8$.

- (A) 2 (B) -1
(C) -2 (D) 3

15. Of a sum of money, A takes $1/3$; then B takes $1/4$ of what is left; then C takes $1/2$ of the new remainder, leaving ₹150. The original sum (in ₹) was:

- (A) 450 (B) 300
(C) 900 (D) 600

16. The sum of four consecutive even integers equals 3 times the largest, decreased by 2. The smallest of them is:

- (A) 6 (B) 10
(C) 4 (D) 8

17. A two-digit number is 6 times its units digit, and its tens digit is 1 less than its units digit. The number is:

- (A) 12 (B) 24
(C) 21 (D) 36

18. Solve: $(2x + 1)/4 - (x - 1)/8 = (5x + 1)/8$.

- (A) -1 (B) 1
(C) 3 (D) 2

19. ₹900 is shared among A, B, C, D so that B gets $1/2$ of A, C gets $1/3$ of A, and D gets $1/4$ of A. A's share (in ₹) is:

- (A) 216 (B) 360
(C) 300 (D) 432

20. Solve: $4 - [2x - \{3 - (1 - x)\}] = 2x - 6$.

- (A) -4 (B) 2
(C) 6 (D) 4

21. A boat goes 24 km downstream and 24 km upstream in a total of 5 hours. If the downstream speed is 12 km/h, the upstream speed (in km/h) is:

- (A) 12 (B) 8
(C) 6 (D) 10

22. How many values of x satisfy $5(x - 2) - 3(x - 4) = 2(x + 1)$?

- (A) exactly one (B) none
(C) infinitely many (every number) (D) exactly two

23. A father's present age is 5 more than 3 times his son's age. Ten years from now the father will be 7 more than twice the son's age then. The son's present age (in years) is:

- (A) 12 (B) 8
(C) 41 (D) 10

24. Solve: $(3x - 2)/(2x - 5) = (3x + 4)/(2x + 1)$.

- (A) 3 (B) 5
(C) -3 (D) -5

25. The denominator of a fraction is 5 more than the numerator. If 4 is added to both, the value becomes $1/2$. The original numerator is:

- (A) 6 (B) 5
(C) 1 (D) 9

26. Solve: $2[3x - (x - \{2x - 4\})] = 5x + 4$.

- (A) 4 (B) -4
(C) 2 (D) 1

27. An angle is 20° more than its complement. The angle measures:

- (A) 35° (B) 70°
(C) 55° (D) 110°

28. ₹720 is divided among A, B, C so that A gets $\frac{3}{4}$ of B and C gets $\frac{5}{4}$ of B. B's share (in ₹) is:

- (A) 240 (B) 180
(C) 300 (D) 320

29. Solve: $(5x + 1)/4 - (3x - 5)/6 = (6x - 5)/12$.

- (A) 6 (B) -6
(C) -3 (D) 3

30. A man is twice as old as his son. Twenty years ago he was four times as old. The man's present age (in years) is:

- (A) 30 (B) 40
(C) 80 (D) 60

31. A vessel holds 90 L of a milk-water mixture in the ratio 7 : 2. How many litres of water must be added to make the ratio 7 : 5?

- (A) 20 (B) 15
(C) 50 (D) 30

32. Solve: $3\{2 - [x - (2x - 5)]\} = 2(x - 3)$.

- (A) -3 (B) 5
(C) 3 (D) 1

33. The sum of the digits of a two-digit number is 12. If the digits are reversed, the number decreases by 18. The original number is:

- (A) 57 (B) 75
(C) 84 (D) 93

34. Solve: $(4x - 4)/3 - (5x - 5)/5 = (4x - 11)/15$.

- (A) -6 (B) 6
(C) -3 (D) 3

35. Pipe A fills a tank in 8 hours, pipe B in 12 hours, and a drain C empties it in 24 hours. With all three open, the time (in hours) to fill the tank is:

- (A) 4 (B) 8
(C) 6 (D) 44

36. For which value of k does $6(x - 1) - 4(x - 3) = 2x + k$ hold for every value of x?

- (A) 0 (B) 6
(C) -6 (D) 2

37. A shopkeeper sold 60 articles; each profitable one earned ₹15 and each loss-making one lost ₹10. If the net gain was ₹400, the number of profitable articles was:

- (A) 40 (B) 20
(C) 44 (D) 16

38. Solve: $7 - \{x - [3 - (2 - x)]\} = 2x + 2$.

- (A) -3 (B) 3
(C) 1 (D) 6

39. A two-digit number has its tens digit equal to 3 times its units digit, and the number exceeds 8 times the units digit by 23. The number is:

- (A) 62 (B) 93
(C) 13 (D) 31

40. Three numbers are such that the second is 4 more than the first and the third is 4 more than the second. If the sum of the first and the third equals twice the second, how many such triples are there?

- (A) exactly one (B) infinitely many
(C) none (D) exactly two

41. A rectangle's length is 3 cm more than twice its breadth. If both length and breadth are increased by 2 cm, the perimeter becomes 50 cm. The original breadth (in cm) is:

- (A) 15 (B) 8
(C) 4 (D) 6

42. How many values of x satisfy $0.25(8x - 4) - 0.5(2x - 6) = 0.1(10x + 20)$?

- (A) exactly one (B) none
(C) exactly two (D) infinitely many (every number)

43. A sum is shared so that for every ₹3 the first child gets, the second gets ₹4 and the third gets ₹5. If the third child gets ₹40 more than the first, the second child's share (in ₹) is:

- (A) 80 (B) 60
(C) 100 (D) 120

44. Solve: $(4x - 5)/4 - (5x - 3)/6 = (x - 4)/12$.

- (A) -5 (B) 3
(C) 5 (D) 7

45. Two stations are 300 km apart. Two trains start toward each other at the same time, one at 50 km/h and the other 20 km/h faster. They meet after a time (in hours) of:

- (A) 6 (B) 2.5
(C) 5 (D) 3

46. If $8x - 5 = 3x + 20$, then the value of $(2x - 1)(x + 2)$ is:

- (A) 5 (B) 9
(C) 45 (D) 63

47. Solve: $2(x - \{3 - [x - (1 - x)]\}) = 3x + 4$.

- (A) 4 (B) -4
(C) 2 (D) 1

48. The angles of a triangle are such that the second is 10° more than the first and the third is twice the first. The largest angle measures:

- (A) 85° (B) 42.5°
(C) 52.5° (D) 95°

49. ₹1000 is shared among A, B, C, D so that each later person gets ₹20 more than the previous one ($B = A + 20$, $C = B + 20$, $D = C + 20$). A's share (in ₹) is:

- (A) 250 (B) 220
(C) 280 (D) 200

50. Solve: $(3x - 4)/4 - (4x - 6)/6 = (2x - 5)/12$.

- (A) -5 (B) 3
(C) 5 (D) 7

51. A man covers $1/3$ of a journey by bus, then $1/2$ of the remaining distance by train, and the last 40 km by car. The total journey (in km) is:

- (A) 120 (B) 80
(C) 60 (D) 240

52. A grandfather's age is 6 less than 7 times his grandson's age. In 4 years the grandfather will be 5 times as old as the grandson then. The grandson's present age (in years) is:

- (A) 8 (B) 71
(C) 11 (D) 10

53. Solve: $x - [x - \{x - (x - 1)\}] = 3 - x$.

- (A) 2 (B) -2
(C) 1 (D) 3

54. A bag has ₹2, ₹5 and ₹10 notes in the ratio 5 : 3 : 2, worth ₹450 in all. The number of ₹10 notes is:

- (A) 50 (B) 30
(C) 20 (D) 10

55. What is the value of $3(2x + 5) - 2(3x - 1)$, and for which x does it hold?

(A) always 17, for every x

(B) 13, only at $x = 0$

(C) 0, only at $x = 1$

(D) depends on x : $12x + 17$

56. A two-digit number is 7 more than 5 times the sum of its digits, and the tens digit exceeds the units digit by 1. The number is:

(A) 23

(B) 43

(C) 21

(D) 32

57. A car travels from P to Q at 40 km/h and returns at 60 km/h, taking 1 hour more on the slower leg. The distance PQ (in km) is:

(A) 60

(B) 120

(C) 240

(D) 100

58. If $5x - 7 = 2x + 8$, then the value of $(x^2 + 5)/(x - 2)$ is:

(A) 5

(B) 25

(C) 10

(D) 30

59. Solve: $(2x + 1)/3 - (3x + 2)/5 = (6x - 6)/15$.

(A) -1

(B) 1

(C) 3

(D) 5

60. If $7x - 4 = 4x + 11$, then the value of $3x^2 - 2x$ is:

(A) 65

(B) 5

(C) 75

(D) 15

Solutions — Mathematics (Class 7), Chapter 15: Finding the Unknown

Marking: +4 correct, -1 wrong, 0 unattempted. Maximum marks **240**.

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	C	21	B	31	D	41	D	51	A
2	A	12	B	22	C	32	C	42	D	52	C
3	D	13	A	23	A	33	B	43	A	53	A
4	D	14	C	24	C	34	A	44	C	54	C
5	D	15	D	25	C	35	C	45	B	55	A
6	B	16	C	26	A	36	B	46	D	56	D
7	D	17	A	27	C	37	A	47	A	57	B
8	B	18	B	28	A	38	B	48	A	58	C
9	D	19	D	29	B	39	D	49	B	59	B
10	C	20	D	30	D	40	B	50	C	60	A

Answer-key distribution: A = 15, B = 15, C = 15, D = 15.

Worked Solutions

1. (B) Work outward. $3 - (x - 4) = 3 - x + 4 = 7 - x$; then $2x - (7 - x) = 2x - 7 + x = 3x - 7$; so $5(3x - 7) - 7 = 15x - 35 - 7 = 15x - 42$. The equation $15x - 42 = 8x + 14$ gives $7x = 56$, so $x = 8$. Substituting back: the bracket equals 17 and $5 \cdot 17 - 7 = 78 = 8 \cdot 8 + 14$. Failing to flip the innermost sign leaves a smaller root near six; a stray negative sign on the constant produces a negative root; leaving the outer bracket un-multiplied by five lands at four.

2. (A) Build outward: $5 - (x - 2) = 7 - x$; then $2x - (7 - x) = 3x - 7$; so $4(3x - 7) = 12x - 28$ and the left side is $9x - (12x - 28) = -3x + 28$. The equation $-3x + 28 = 3x - 8$ gives $36 = 6x$, so $x = 6$. Check: the bracket value is 11, $4 \cdot 11 = 44$, $9 \cdot 6 - 44 = 10$ and $3 \cdot 6 - 8 = 10$. Mishandling the subtracted bracket doubles the apparent root to twelve; a sign slip gives a negative; not distributing the four lands at four.

3. (D) Inside: $x - (3 - 2x) = x - 3 + 2x = 3x - 3$, so $2(3x - 3) = 6x - 6$, and the left side is $(6x - 3) - (6x - 6) = 3$, a constant. The equation $3 = x + 5$ gives $x = -2$. Check at $x = -2$: $3(-5) - 2[-2 - (3 + 4)] = -15 + 18 = 3$ and $-2 + 5 = 3$. Dropping a sign on the double bracket makes the x-terms fail to cancel and yields two or one; reading the right side as $x - 5$ gives eight.

4. (D) Let the number be x . Then $4x - 9 = 2x + 7$, so $2x = 16$ and $x = 8$. Check: $4 \cdot 8 - 9 = 23$ and $2 \cdot 8 + 7 = 23$. Mis-signing the subtraction gives a negative; dividing sixteen by sixteen instead of two gives one; reporting the intermediate $2x = 16$ gives sixteen.

5. (D) Multiply every term by the LCD 12: $3(3x - 1) - 2(2x + 3) = x + 11$. Expand: $9x - 3 - 4x - 6 = x + 11$, so $5x - 9 = x + 11$, giving $4x = 20$ and $x = 5$. Check: $(14)/4 - (13)/6 = 3.5 - 2.1666... = 1.3333...$ and $(16)/12 = 1.3333...$. A sign error on $-2(2x + 3)$ shifts the root negative; clearing only one denominator gives three or seven.

6. (B) After spending one-fourth he keeps three-fourths; spending one-third of that keeps two-thirds of three-fourths, i.e. one-half of the original. So half the money is ₹360, and the original was ₹720. Check: spends 180 leaving 540, then spends 180 leaving 360. Stopping after the first step reports the 540 remainder; treating both fractions as taken from the original ($7/12$ spent) gives the wrong 480-ish value; 900 is a multiplication slip.

7. (D) Let the integers be n , $n + 1$, $n + 2$. The first plus the third is $2n + 2$, and this equals the second plus four: $2n + 2 = (n + 1) + 4$. So $n = 3$ and the middle integer is $n + 1 = 4$. Check: $3 + 5 = 8 = 4 + 4$. Reporting n itself gives three; reporting the largest gives five; an off-by-one in the set-up gives two.

8. (B) Let the son be s , so the father is $60 - s$. In twelve years the son is $s + 12$ and the father is $72 - s$, with $72 - s = 2(s + 12)$. Expand: $72 - s = 2s + 24$, so $48 = 3s$ and $s = 16$. Check: son 16, father 44; in twelve years 28 and 56, and $56 = 2 \times 28$. Reporting the father's present age gives 44; mishandling the added twelve gives twenty; a division slip gives twelve.

9. (D) Multiply every term by the LCD 6: $2(3x - 1) - (x + 2) = 6x - 10$. Expand: $6x - 2 - x - 2 = 6x - 10$, so $5x - 4 = 6x - 10$, giving $6 = x$. Check: $(17)/3 - (8)/6 = 5.6666... - 1.3333... = 4.3333...$ and $(26)/6 = 4.3333...$. A dropped minus on $(x + 2)$ flips the sign; clearing one denominator only gives three or two.

10. (C) Let the one-way distance be d . Then $d/12 + d/18 = 5$; multiplying by 36 gives $3d + 2d = 180$, so $5d = 180$ and $d = 36$. Check: $36/12 = 3$ h and $36/18 = 2$ h, total 5 h. Averaging the two speeds to fifteen and dividing 75 by fifteen wrongly suggests 75; thirty and forty-five are arithmetic slips.

11. (C) Let the daughter be d , so the mother is $4d$. Eight years ago: $4d - 8 = 8(d - 8)$. Expand: $4d - 8 = 8d - 64$, so $56 = 4d$ and $d = 14$; the mother is $4 \times 14 = 56$. Check: eight years ago 6 and 48, and $48 = 8 \times 6$. Reporting the daughter gives fourteen; reporting the mother's past age gives forty-eight; a constant slip gives sixty-four.

12. (B) Multiply every term by the LCD 6: $2(4x - 1) - (5x - 1) = 4x - 9$. Expand: $8x - 2 - 5x + 1 = 4x - 9$, so $3x - 1 = 4x - 9$, giving $8 = x$. Check: $(31)/3 - (39)/6 = 10.3333... - 6.5 = 3.8333...$ and $(23)/6 = 3.8333...$. Distributing $-(5x - 1)$ as $-5x - 1$ corrupts the constant and shifts the root; clearing one denominator gives four or two.

13. (A) Let the number be n . Then $(2/3)n = (1/2)n + 4$, so $(2/3 - 1/2)n = 4$, i.e. $(1/6)n = 4$ and $n = 24$. Check: two-thirds of 24 is 16, half is 12, and $16 = 12 + 4$. Doubling the gap gives forty-eight; treating the difference as one-third gives twelve; treating it as one-half gives eight.

14. (C) Multiply every term by the LCD 8: $2(4x + 1) - (3x - 3) = 6x + 7$. Expand: $8x + 2 - 3x + 3 = 6x + 7$, so $5x + 5 = 6x + 7$, giving $-2 = x$. Check: $(-7)/4 - (-9)/8 = -1.75 + 1.125 = -0.625$ and $(-5)/8 = -0.625$. A sign slip on $-(3x - 3)$ flips the answer to positive two; mis-clearing gives -1 or three.

15. (D) After A two-thirds remain; B takes one-fourth of that, leaving three-fourths of two-thirds, which is one-half of the sum; C takes one-half of that, leaving one-fourth of the sum. So one-fourth equals ₹150 and the sum is ₹600. Check: A 200 leaves 400, B 100 leaves 300, C 150 leaves 150. Stopping a step early gives 300 or 450; over-counting the fractions gives 900.

16. (C) Let them be $n, n + 2, n + 4, n + 6$. Their sum is $4n + 12$, and three times the largest minus two is $3(n + 6) - 2 = 3n + 16$. Setting $4n + 12 = 3n + 16$ gives $n = 4$, so the integers are 4, 6, 8, 10 and the smallest is 4. Check: sum $28 = 3 \times 10 - 2$. Reporting the second term gives six; the largest gives ten; the third gives eight.

17. (A) Let the units digit be u and tens digit $u - 1$, number $10(u - 1) + u = 11u - 10$. The condition $11u - 10 = 6u$ gives $5u = 10$, so $u = 2$, tens digit 1, number 12. Check: $6 \times 2 = 12$ and $1 = 2 - 1$. Reversing the digit relation gives twenty-one; ignoring the units-digit link gives twenty-four or thirty-six.

18. (B) Multiply every term by the LCD 8: $2(2x + 1) - (x - 1) = 5x + 1$. Expand: $4x + 2 - x + 1 = 5x + 1$, so $3x + 3 = 5x + 1$, giving $2 = 2x$ and $x = 1$. Check: $(3)/4 - (0)/8 = 0.75$ and $(6)/8 = 0.75$. A sign slip on $-(x - 1)$ gives -1 ; mis-clearing the denominators gives three or two.

19. (D) Let A's share be a . Then $a + a/2 + a/3 + a/4 = 900$. Over the LCD 12 this is $(12a + 6a + 4a + 3a)/12 = 25a/12 = 900$, so $a = 432$. Check: $432 + 216 + 144 + 108 = 900$. Using only B's half gives 216; mis-summing the fractions gives 360 or 300.

20. (D) Innermost: $3 - (1 - x) = 2 + x$; then $2x - (2 + x) = x - 2$; then $4 - (x - 2) = 6 - x$. The equation $6 - x = 2x - 6$ gives $12 = 3x$, so $x = 4$. Check: $4 - [8 - \{3 + 3\}] = 4 - [8 - 6] = 2$ and $2 \times 4 - 6 = 2$. A flipped nested sign gives a negative root or two; not distributing the outer subtraction gives six.

21. (B) Downstream time is $24/12 = 2$ hours, so the upstream leg takes $5 - 2 = 3$ hours, giving an upstream speed of $24/3 = 8$ km/h. Check: $2 + 3 = 5$ hours. Taking the same speed both ways gives twelve; mis-dividing gives six; averaging the legs gives ten.

22. (C) Expand: $5x - 10 - 3x + 12 = 2x + 2$, so the left side is $2x + 2$ and equals the right side $2x + 2$ for every x — infinitely many solutions. Solving and reading off a single root suggests 'exactly one'; reading the equal constants as unequal suggests 'none'; a linear equation can never have 'exactly two'.

23. (A) Let the son be s , so the father is $3s + 5$. In ten years: $3s + 15 = 2(s + 10) + 7 = 2s + 27$, so $s = 12$. Check: father now 41; in ten years son 22, father 51, and $2 \times 22 + 7 = 51$. Reporting the father's present age gives 41; mishandling the added ten gives eight or ten.

24. (C) Cross-multiply: $(3x - 2)(2x + 1) = (3x + 4)(2x - 5)$. Left: $6x^2 + 3x - 4x - 2 = 6x^2 - x - 2$. Right: $6x^2 - 15x + 8x - 20 = 6x^2 - 7x - 20$. The $6x^2$ cancels: $-x - 2 = -7x - 20$, so $6x = -18$ and $x = -3$. Check: both sides equal $(-11)/(-11) = 1$. A dropped minus gives positive three; transposition slips give five or negative five.

25. (C) Let the numerator be n and denominator $n + 5$. After adding four: $(n + 4)/(n + 9) = 1/2$, so $2(n + 4) = n + 9$, giving $2n + 8 = n + 9$ and $n = 1$. Check: original $1/6$ becomes $5/10 = 1/2$. Reporting the denominator gives six; off-by-one set-ups give five or nine.

26. (A) Innermost: $x - (2x - 4) = -x + 4$; then $3x - (-x + 4) = 4x - 4$; doubling gives $8x - 8$. The equation $8x - 8 = 5x + 4$ gives $3x = 12$, so $x = 4$. Check: the bracket is $4 \cdot 4 - 4 = 12$, $2 \times 12 = 24$ and $5 \times 4 + 4 = 24$. A nested-sign error gives a negative root or two; a constant slip gives one.

27. (C) Let the angle be x ; its complement is $90 - x$, and $x = (90 - x) + 20$. So $2x = 110$ and $x = 55$. Check: complement 35, and $55 = 35 + 20$. The complement 35 is a trap; using the supplement instead of the complement gives 70 or 110.

28. (A) Let $B = b$. Then $A = (3/4)b$ and $C = (5/4)b$, so the total is $(3/4 + 1 + 5/4)b = 3b = 720$, giving $b = 240$. Check: $A = 180$, $B = 240$, $C = 300$, total 720. Reporting A gives 180; reporting C gives 300; a slip gives 320.

29. (B) Multiply every term by the LCD 12: $3(5x + 1) - 2(3x - 5) = 6x - 5$. Expand: $15x + 3 - 6x + 10 = 6x - 5$, so $9x + 13 = 6x - 5$, giving $3x = -18$ and $x = -6$. Check: $(-29)/4 - (-23)/6 = -7.25 + 3.8333... = -3.4166...$ and $(-41)/12 = -3.4166...$. A sign slip on $-2(3x - 5)$ flips the root to positive; mis-clearing gives -3 or 3 .

30. (D) Let the son be s and the man $2s$. Twenty years ago: $2s - 20 = 4(s - 20)$. Expand: $2s - 20 = 4s - 80$, so $60 = 2s$ and $s = 30$; the man is $2 \times 30 = 60$. Check: twenty years ago 10 and 40, and $40 = 4 \times 10$. Reporting the son gives thirty; reporting the man's past age gives forty; doubling wrongly gives eighty.

31. (D) Milk is $(7/9) \times 90 = 70$ L and water $(2/9) \times 90 = 20$ L. Adding water keeps milk at 70 L; for $7 : 5$ the water must be $(5/7) \times 70 = 50$ L, so the amount added is $50 - 20 = 30$ L. Check: $70 : 50 = 7 : 5$. Re-stating the original water gives twenty; halving the gap gives fifteen; reporting the final water total gives fifty.

32. (C) Innermost: $x - (2x - 5) = -x + 5$; then $2 - (-x + 5) = x - 3$; tripling gives $3x - 9$. The equation $3x - 9 = 2x - 6$ gives $x = 3$. Check: $3\{2 - [3 - 1]\} = 3 \times 0 = 0$ and $2(3 - 3) = 0$. A nested-sign flip gives a negative root or five; a constant slip gives one.

33. (B) Let tens t and units u with $t + u = 12$. Reversing gives $10u + t = (10t + u) - 18$, so $9u - 9t = -18$ and $t - u = 2$. With $t + u = 12$, $t = 7$ and $u = 5$, so the number is 75. Check: reversed $57 = 75 - 18$ and $7 + 5 = 12$. The reversed 57 is a trap; 84 and 93 meet the digit sum but not the decrease-by-18 condition.

34. (A) Multiply every term by the LCD 15: $5(4x - 4) - 3(5x - 5) = 4x - 11$. Expand: $20x - 20 - 15x + 15 = 4x - 11$, so $5x - 5 = 4x - 11$, giving $x = -6$. Check: $(-28)/3 - (-35)/5 = -9.3333... + 7 = -2.3333...$ and $(-35)/15 = -2.3333...$. A sign slip on $-3(5x - 5)$ flips the root; mis-clearing gives -3 or 3 .

35. (C) In one hour A fills $1/8$, B fills $1/12$, C drains $1/24$; the net fill rate is $1/8 + 1/12 - 1/24 = 3/24 + 2/24 - 1/24 = 4/24 = 1/6$ of the tank, so it fills in 6 hours. Ignoring the drain (rate $5/24$) drifts toward 4; averaging the times gives 8; adding the times gives 44.

36. (B) Expand the left: $6x - 6 - 4x + 12 = 2x + 6$. The identity $2x + 6 = 2x + k$ holds for all x only when $k = 6$. Forgetting the $+12$ gives 0; a sign error gives -6 or 2 .

37. (A) Let profitable articles be p , so loss articles number $60 - p$. Then $15p - 10(60 - p) = 400$, i.e. $25p - 600 = 400$, so $25p = 1000$ and $p = 40$. Check: 40 earn ₹600 and 20 lose ₹200, net ₹400. The complement 20 is a swap trap; sign errors give 44 or 16.

38. (B) Innermost: $3 - (2 - x) = 1 + x$; then $x - (1 + x) = -1$; then $7 - (-1) = 8$, a constant. The equation $8 = 2x + 2$ gives $2x = 6$, so $x = 3$. Check: the left side is the constant 8 and $2 \times 3 + 2 = 8$. Losing a nested sign so the x -terms fail to cancel gives a negative or one; a slip gives six.

39. (D) Let the units digit be u and the tens digit $3u$, so the number is $10(3u) + u = 31u$. The condition $31u = 8u + 23$ gives $23u = 23$, so $u = 1$, tens digit 3, number 31. Check: $31 = 8 \times 1 + 23$. Doubling u gives 62; tripling gives 93; reversing the digit roles gives 13.

40. (B) Let the first be n ; the second is $n + 4$ and the third $n + 8$. The first plus the third is $2n + 8$, and twice the second is $2(n + 4) = 2n + 8$ — equal for every n . So any starting value works: infinitely many triples. Finding one example suggests 'exactly one'; reading the constants as unequal suggests 'none'; a single linear identity never gives 'exactly two'.

41. (D) Let the breadth be b , so the length is $2b + 3$. The new dimensions are $(2b + 5)$ and $(b + 2)$, with perimeter $2[(2b + 5) + (b + 2)] = 2(3b + 7) = 50$. So $3b + 7 = 25$, giving $3b = 18$ and $b = 6$. Check: original 15 by 6, new 17 by 8, perimeter 50. Reporting the length gives 15; mishandling the added 2's gives 8 or 4.

42. (D) Expand: $2x - 1 - x + 3 = x + 2$, so the left side is $x + 2$ and equals the right side $x + 2$ for every x — infinitely many solutions. Treating $0.5(2x - 6)$ as $x - 6$ makes the x -terms appear unequal, suggesting 'exactly one'; reading the constants as different suggests 'none'; a linear identity is never 'exactly two'.

43. (A) Let the common part be k , so the shares are $3k$, $4k$, $5k$. The third exceeds the first by $5k - 3k = 2k = 40$, so $k = 20$ and the second child gets $4k = 80$. Check: $60, 80, 100$, and $100 - 60 = 40$. Reporting the first gives 60 ; the third gives 100 ; the total drifts toward 120 .

44. (C) Multiply every term by the LCD 12 : $3(4x - 5) - 2(5x - 3) = x - 4$. Expand: $12x - 15 - 10x + 6 = x - 4$, so $2x - 9 = x - 4$, giving $x = 5$. Check: $(15)/4 - (22)/6 = 3.75 - 3.6666... = 0.0833...$ and $(1)/12 = 0.0833...$. A sign slip on $-2(5x - 3)$ shifts the root; clearing one denominator gives 3 or 7 .

45. (B) The faster train runs at $50 + 20 = 70$ km/h, so the closing speed is $50 + 70 = 120$ km/h. The 300 km gap closes in $300/120 = 2.5$ hours. Using only one speed gives $300/50 = 6$; using the difference 20 km/h gives an inflated time; 5 and 3 are arithmetic slips.

46. (D) Solve: $8x - 5 = 3x + 20$ gives $5x = 25$, so $x = 5$. Then $(2x - 1)(x + 2) = (9)(7) = 63$. Stopping at $x = 5$ reports 5 ; computing only one factor gives 9 ; a multiplication slip gives 45 .

47. (A) Innermost: $x - (1 - x) = 2x - 1$; then $3 - (2x - 1) = 4 - 2x$; then $x - (4 - 2x) = 3x - 4$; doubling gives $6x - 8$. The equation $6x - 8 = 3x + 4$ gives $3x = 12$, so $x = 4$. Check: the bracket equals $3 \cdot 4 - 4 = 8$, $2 \times 8 = 16$ and $3 \times 4 + 4 = 16$. A nested-sign error gives a negative or two; a slip gives one.

48. (A) Let the first angle be x ; the second is $x + 10$ and the third $2x$. Their sum is $4x + 10 = 180$, so $x = 42.5$ and the angles are 42.5° , 52.5° , 85° ; the largest is the third, 85° . Check: $42.5 + 52.5 + 85 = 180$. Reporting the first angle gives 42.5° , the second 52.5° ; a slip gives 95° .

49. (B) Let $A = a$, so $B = a + 20$, $C = a + 40$, $D = a + 60$. Their sum is $4a + 120 = 1000$, giving $4a = 880$ and $a = 220$. Check: $220, 240, 260, 280$ sum to 1000 . Splitting equally gives 250 ; reporting D gives 280 ; a slip gives 200 .

50. (C) Multiply every term by the LCD 12 : $3(3x - 4) - 2(4x - 6) = 2x - 5$. Expand: $9x - 12 - 8x + 12 = 2x - 5$, so $x = 2x - 5$, giving $x = 5$. Check: $(11)/4 - (14)/6 = 2.75 - 2.3333... = 0.4166...$ and $(5)/12 = 0.4166...$. A sign slip on $-2(4x - 6)$ flips the root negative; clearing one denominator only gives 3 or 7 .

51. (A) Let the total be D . The bus covers $(1/3)D$, leaving $(2/3)D$; the train covers half of that, $(1/3)D$, leaving the other half $(1/3)D = 40$ km for the car. So $(1/3)D = 40$ and $D = 120$. Check: bus 40 , train 40 , car 40 . Stopping after one stage gives 80 ; mis-reading the remainder gives 60 ; doubling gives 240 .

52. (C) Let the grandson be g , so the grandfather is $7g - 6$. In four years: $7g - 2 = 5(g + 4) = 5g + 20$, so $2g = 22$ and $g = 11$. Check: grandfather now 71; in four years grandson 15, grandfather 75, and $75 = 5 \times 15$. Reporting the grandfather's age gives 71; mishandling the added 4 gives 8 or 10.

53. (A) Work outward: $x - 1$; then $x - (x - 1) = 1$; then $x - \{1\} = x - 1$; then $x - (x - 1) = 1$, so the whole left side is the constant 1. The equation $1 = 3 - x$ gives $x = 2$. Check: every nested layer collapses to 1 and $3 - 2 = 1$. Mis-tracking a sign layer gives a negative or one; mis-evaluating gives three.

54. (C) Let the counts be $5k$, $3k$, $2k$. The value is $2(5k) + 5(3k) + 10(2k) = 10k + 15k + 20k = 45k = 450$, so $k = 10$ and the ₹10 notes number $2k = 20$. Check: $50 + 150 + 200 = ₹450$ with 50, 30, 20 notes. Reading the ₹2 count gives 50, the ₹5 count gives 30; halving k gives 10.

55. (A) Expand: $6x + 15 - 6x + 2 = 17$. The x -terms cancel, so the expression is the constant 17 for every value of x . Misreading $-2(3x - 1)$ as $-6x - 2$ gives 13; dropping both constants gives 0; failing to notice that the x -terms cancel gives $12x + 17$.

56. (D) Let the units digit be u and the tens digit $u + 1$, so the number is $10(u + 1) + u = 11u + 10$ and the digit sum is $2u + 1$. The condition $11u + 10 = 5(2u + 1) + 7 = 10u + 12$ gives $u = 2$, tens digit 3, number 32. Check: $5 \times 5 + 7 = 32$ and $3 = 2 + 1$. Reversing the digit relation gives 23; choosing $u = 3$ gives 43; a slip gives 21.

57. (B) Let $PQ = d$. Then the slow leg takes $d/40$ and the fast leg $d/60$, with $d/40 = d/60 + 1$. Multiplying by 120: $3d = 2d + 120$, so $d = 120$. Check: $120/40 = 3$ h and $120/60 = 2$ h, a one-hour difference. Halving gives 60; doubling gives 240; an arithmetic slip gives 100.

58. (C) Solve: $5x - 7 = 2x + 8$ gives $3x = 15$, so $x = 5$. Then $(x^2 + 5)/(x - 2) = (25 + 5)/(5 - 2) = 30/3 = 10$. Stopping at $x = 5$ reports 5; computing only x^2 gives 25; forgetting to divide by $(x - 2)$ gives 30.

59. (B) Multiply every term by the LCD 15: $5(2x + 1) - 3(3x + 2) = 6x - 6$. Expand: $10x + 5 - 9x - 6 = 6x - 6$, so $x - 1 = 6x - 6$, giving $5 = 5x$ and $x = 1$. Check: $(3)/3 - (5)/5 = 1 - 1 = 0$ and $(0)/15 = 0$. A sign slip on $-3(3x + 2)$ shifts the root; clearing one denominator only gives 3 or 5.

60. (A) Solve: $7x - 4 = 4x + 11$ gives $3x = 15$, so $x = 5$. Then $3x^2 - 2x = 3(25) - 10 = 75 - 10 = 65$. Stopping at $x = 5$ reports 5; computing only $3x^2$ gives 75; using $3x - 2x = x$ gives 15.